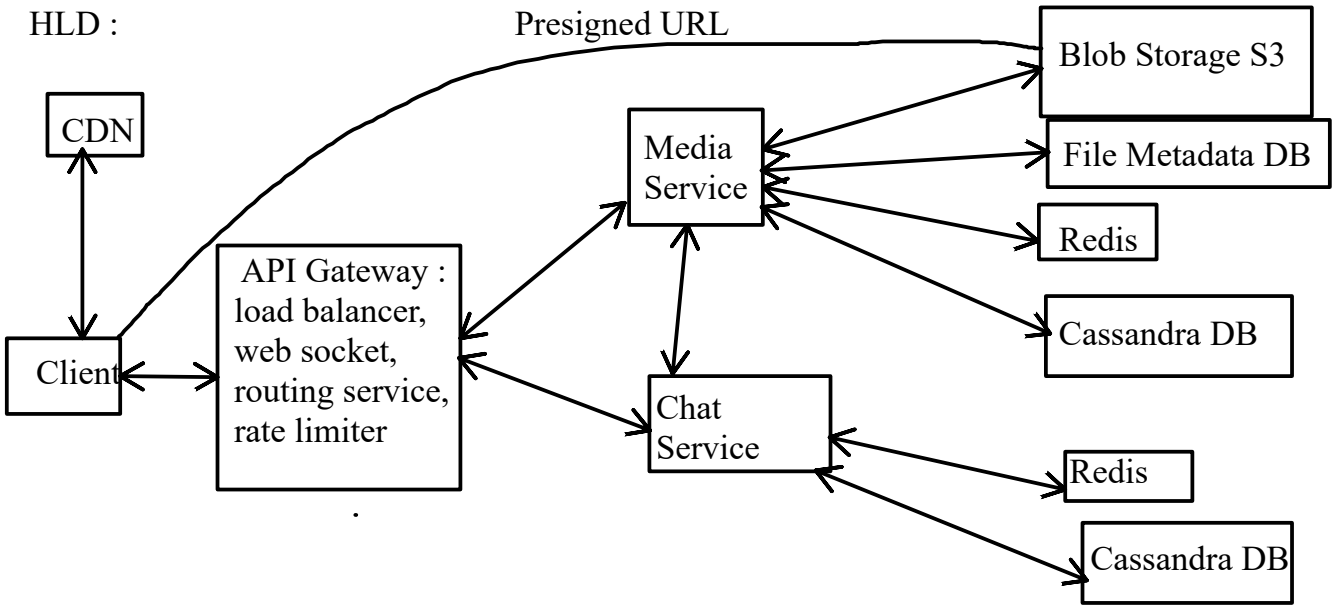
Whatsapp

Functional requirements:
user should be able to share media (Images/videos) based on his preference,
user should be able to search for chat among the existing chat based on his preference,
user should be able to share any filea above>50GB based on his preference,

Non-functional requirements:
data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:
user,
chat,
messages,
media

API :
POST:/upload
{
userID,
{filemetadata}
}
GET:/search/{chatID}

HLD :